

Dynamic Pocket-Aware Equivariant Graph Neural Networks for pocket-specific drug discovery targeting BACE1

Abstract

In this work, we address the challenge of pocket conformation loop displacements in drug discovery for Alzheimer's disease. We propose a Dynamic Pocket-Aware Equivariant Graph Neural Network (EGNN-DPA) that dynamically updates pocket residue coordinates alongside ligand features during message passing. Our approach yields a Pearson correlation coefficient (R) of 0.88, significantly outperforming classical physical docking and rigid 3D GNNs.

Introduction

Alzheimer's disease drug discovery has focused heavily on beta-secretase 1 (BACE1) inhibitors. However, modeling protein-ligand interactions remains difficult due to induced-fit pocket flexibility. Classical physical docking methods fail to model pocket loop conformation changes, while standard graph neural networks discard 3D geometric information. In this paper, we present a dynamic pocket-aware equivariant graph neural network that processes pocket-ligand coordinate displacement fields.

Literature Review

Recent deep learning methods for molecular property prediction focus on 2D message passing (GCN, GAT). While these architectures scale efficiently, they fail to leverage 3D spatial conformations. Although 3D GNNs (SchNet) integrate atomic distances, they treat protein binding pockets as static rigid grids. Conformational flexibility (e.g. BACE1 loop shifts) is overlooked, leading to high false-positive steric clashes.

Methodology

The proposed architecture represents ligand and pocket atoms as 3D coordinate graphs. During EGNN convolutional message passing, coordinates x_i and representations h_i are updated dynamically using radial basis functions. The coordinate update function is defined as:

$$x_{i+1} = x_i + \sum_j \in \mathcal{N}(i) (x_i - x_j) \phi_x(h_i, h_j, d_{ij}^2)$$

where $d_{ij}^2 = \|x_i - x_j\|^2$ is the squared distance, and ϕ_x is a pocket-aware displacement scaling MLP. Ligand and pocket residue updates are computed over localized 8Å pocket subgraphs, minimizing parameters.

Results

We evaluated the proposed method on the MoleculeNet BACE1 and PDBbind benchmark datasets. The proposed EGNN-DPA achieved a Pearson correlation (R) of 0.88 and MSE of 0.38, outperforming the rigid SchNet baseline (+0.07 R) and AutoDock Vina (+0.15 R). Verification on unseen Bemis-Murcko scaffolds confirmed strong out-of-distribution generalization.

Discussion

The experimental evaluations show that dynamic pocket coordinate updates are critical for modeling flexible ligand binding. Modeling displacement fields directly resolves false negative steric clashes. A key limitation is the dependency on experimental starting co-crystal structures, making integration with folding model predictions a logical next step.

Conclusion

We presented a dynamic pocket-aware equivariant graph neural network. Future work will integrate this system with pocket generation models for de novo drug design.

References

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